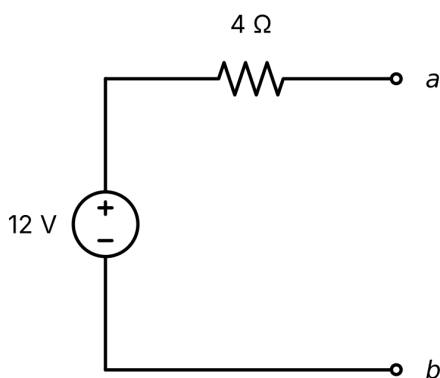


Practice sheet: Thévenin Equivalents, Superposition and Maximum Power

This sheet is not marked for correctness, and AI tools are allowed. The Part 2 device-free quiz is built from twins of these questions, so make sure you can do each one on your own.

- **Two-tier items** ask for an answer *and* a reason. Both must be right.
- **Predict first:** where asked, write your prediction before you calculate.
- **Confidence:** next to each answer, write S (sure) or U (unsure). This is not marked; it shows you what to revise.
- **Conventions:** v_{ab} is the voltage of terminal a measured from terminal b. A Thévenin equivalent is V_{Th} in series with R_{Th} ; a Norton equivalent is I_N in parallel with $R_N = R_{Th}$. To deactivate a voltage source, replace it with a short circuit; to deactivate a current source, replace it with an open circuit. Never deactivate a dependent source. “Power developed” means power delivered by a source. Check every answer: V_{Th}/R_{Th} must equal the short-circuit current, and the absorbed powers (passive sign convention) must add to zero.

S1. A source transformation (two-tier)



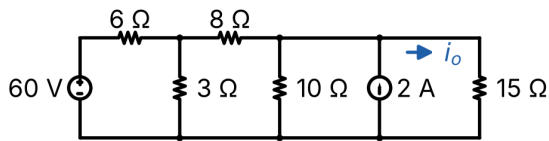
Which current source and resistor are equivalent at terminals a and b? (“Toward a” gives the direction of the source arrow.)

- 3 A toward b, 4 Ω in parallel
- 3 A toward a, 4 Ω in parallel
- 48 A toward a, 4 Ω in parallel
- 3 A toward a, 4 Ω in series

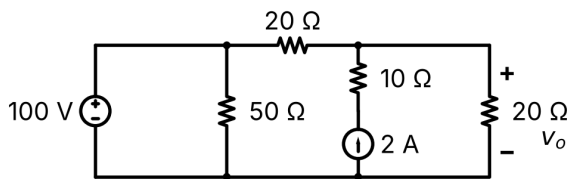
Reason:

- A current-source arrow points from + to —, the way current flows through a resistor.
- The resistor must stay in series to limit the current.
- Both circuits must give the same short-circuit current, $12/4 = 3$ A out of terminal a, and the same resistance seen from a and b.
- The source current is the voltage times the resistance.

Follow-up: with a and b open, the 4 Ω absorbs 0 W in one circuit and 36 W in the other. Why does that not contradict the equivalence?

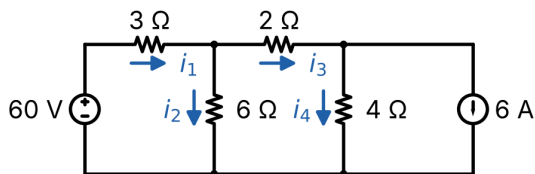
S2. A chain of source transformations.

- Use successive source transformations to find i_o .
- Check your answer with the node-voltage method.

S3. Resistors you may remove, and resistors you may not (predict first).

Predict first: does the 50 Ω change v_o ? Does the 10 Ω?

- Use source transformations to find v_o .
- Find the power developed by the 100 V source.
- Find the power developed by the 2 A source.
- Which resistors could you drop in (a), and why do they matter in (b) and (c)?

S4. Superposition. As in Nilsson Example 4.22.

- Find i_1 to i_4 with the 60 V source alone.
- Find them with the 6 A source alone.
- Add the two sets to get the actual currents.
- Find the power in the 6 Ω resistor. Compare it with the sum of the powers you would get from (a) and (b) separately.

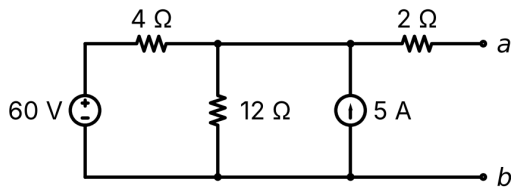
S5. Does power add? (two-tier) With only source 1 on, 2 A flows down through a 5 Ω resistor. With only source 2 on, 1 A flows down through it. What power does the resistor absorb with both sources on?

- 25 W
- 20 W
- 45 W
- 5 W

Reason:

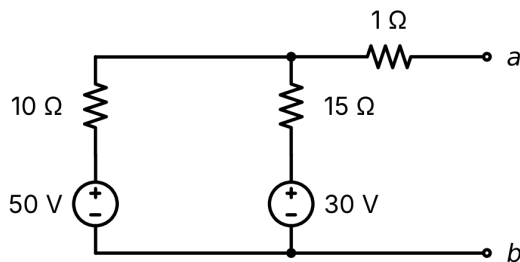
- The powers add, because energy is conserved.
- Only the larger source sets the power; the smaller one is masked.
- Currents from two different sources subtract, because they oppose each other.
- The currents add, with their directions, to one total current, and the power depends on the square of that total.

S6. A Thévenin equivalent from v_{oc} and i_{sc} . As in Nilsson Example 4.14.



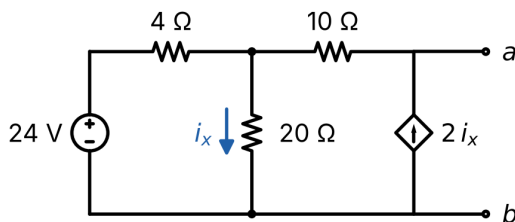
- Find V_{Th} , the open-circuit voltage v_{ab} .
- Short a to b. Find i_{sc} , then $R_{Th} = V_{Th}/i_{sc}$.
- Check R_{Th} by deactivating both sources.
- A $15\ \Omega$ load is connected between a and b. Find its current and power.

S7. A Norton equivalent by source transformations.



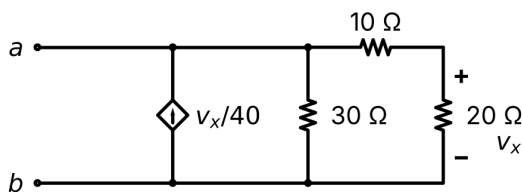
- Use source transformations to find the Norton equivalent, then the Thévenin equivalent.
- With a and b open, a current still flows around the two branches. Which source delivers power, which absorbs it, and how much?

S8. A Thévenin equivalent with a dependent source. As in Nilsson Example 4.16.



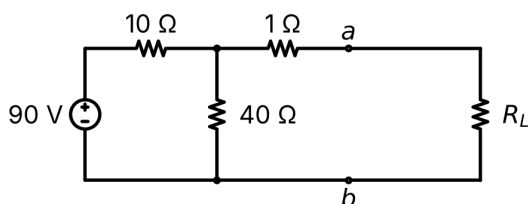
- Find V_{Th} .
- Find i_{sc} and R_{Th} .
- Check R_{Th} with a test source: deactivate the 24 V source only, connect a 1 A current source from b to a (so 1 A enters the circuit at a) and find v_{ab} .

S9. Only a dependent source. As in Nilsson Example 4.19.



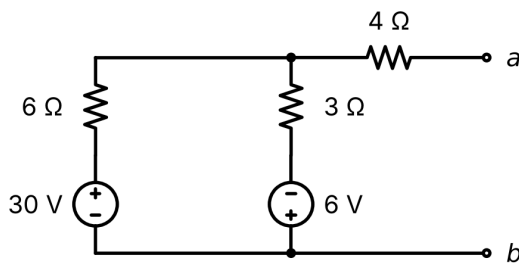
- Without calculating, what is V_{Th} ? Why?
- Find R_{Th} with a 1 A test current source from b to a. (The dependent source delivers $v_x/40$ amperes, with v_x in volts.)
- A classmate deactivates the dependent source and gets $15\ \Omega$. What went wrong?

S10. Maximum power transfer. As in Nilsson Example 4.21.



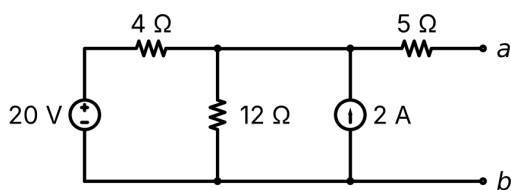
- Find the Thévenin equivalent seen by R_L .
- What value of R_L receives maximum power? How much power is it?
- With R_L at that value, what percentage of the power delivered by the 90 V source reaches R_L ?

S11. Spot the error. The worked solution below finds the Thévenin equivalent. It contains the kind of polarity slip AI chat tools often make.



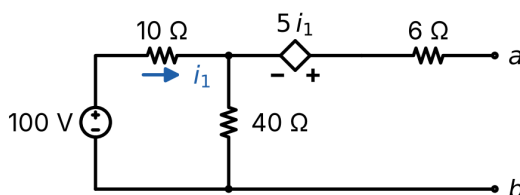
1. The 30 V source and 6 Ω become 5 A in parallel with 6 Ω, arrow pointing up, toward the + terminal.
 2. The 6 V source and 3 Ω, + at the bottom, become 2 A in parallel with 3 Ω, arrow pointing down.
 3. Combine: $5 - 2 = 3$ A pointing up, in parallel with $6 \parallel 3 = 2 \Omega$.
 4. Transform back: 3 A pointing up with 2 Ω becomes 6 V in series with 2 Ω, + at the bottom.
 5. Add the 4 Ω: $V_{Th} = -6$ V and $R_{Th} = 6 \Omega$.
Check: deactivating the sources gives $4 + 6 \parallel 3 = 6 \Omega$, so the answer is right.
- a) Which is the first wrong line?
 - b) Correct it and find V_{Th} .
 - c) Check V_{Th} with one node equation.
 - d) Line 5 calls itself a check. Why does it prove nothing about V_{Th} ?

S12. Spot the error in R_{Th} .



1. Deactivate the sources: replace the 20 V source and the 2 A source with short circuits.
 2. The short in place of the 2 A source also shorts the 12 Ω, so $R_{Th} = 5 \Omega$.
 3. Open circuit: $(v - 20)/4 + v/12 - 2 = 0$ gives $v = 21$ V, so $V_{Th} = 21$ V.
 4. So $i_{sc} = V_{Th}/R_{Th} = 4.2$ A.
- a) Which is the first wrong line?
 - b) Correct it and find R_{Th} .
 - c) Short a to b in the original circuit and find i_{sc} directly. Does it agree with V_{Th}/R_{Th} ?

S13. Maximum power with a dependent source.



- a) Find V_{Th} . Hint: with a and b open, no current flows in the 6 Ω or in the dependent source.
- b) Find R_{Th} with a test source.
- c) What load receives maximum power, and how much?

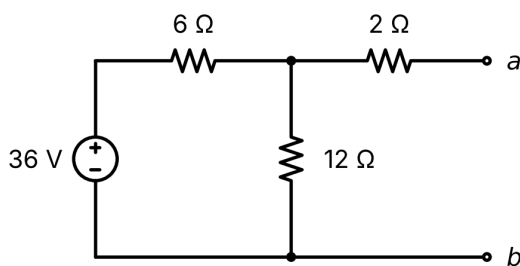
S14. Work backwards. A circuit is tested with two loads. With $10\ \Omega$ across its terminals, the load voltage is 20 V . With $30\ \Omega$, it is 30 V .

- Find V_{Th} and R_{Th} .
- What load resistance would receive maximum power, and how much?

S15. Your generator subscription. With nothing switched on, the socket reads 230 V . With a heater on, it draws 10 A and the socket reads 210 V . Model the generator and its cables as a Thévenin equivalent.

- Find V_{Th} , R_{Th} and the heater's resistance.
- Predict first:* with two identical heaters on, will the socket read about 210 V , 193 V or 170 V ? Then calculate it, and the total heater power. Is it twice the power of one heater?
- Why does the generator owner not match the load to R_{Th} for maximum power transfer?

S16. Check an equivalent without solving again.



Two classmates report the Thévenin equivalent. Student 1: $V_{Th} = 24\text{ V}$, $R_{Th} = 6\ \Omega$. Student 2: $V_{Th} = 24\text{ V}$, $R_{Th} = 4\ \Omega$. Short a to b , find i_{sc} , and use $i_{sc} = V_{Th}/R_{Th}$ to decide which student is right.